

Stop Believing in Haircuts*

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Abstract

We investigate how frictions in euro area repo markets distort the setting of haircuts — often presented as the main risk management tool in repo transactions. Using unique European repo data, we show that balance sheet constraints, collateral heterogeneity, and dealer market power lead to suboptimal haircuts from a risk-management perspective, especially in collateral-driven repos. We develop a theoretical model explaining these distortions and their effects on market outcomes. Our findings highlight the underlying frictions result in more volatile repo rates and volumes, and have important implications for monetary policy and financial stability.

KEYWORDS: REPO MARKETS, HAIRCUTS, FRICTIONS, BALANCE SHEET CONSTRAINTS, MONETARY POLICY TRANSMISSION.

JEL CLASSIFICATION: E43, E44, E52.

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Non-technical Summary

This paper studies how haircuts are set in euro area repurchase agreement (repo) markets and why, in practice, they often do not function as standard risk-management tools.

Repos are the main instruments in short-term money markets and consist of loans secured by government bonds. A distinctive feature of repos is that they can be used either to obtain cash (funding-driven repos) or to obtain specific securities (collateral-driven repos). In principle, haircuts — the difference between the value of the collateral and the cash loan amount — should be positive in funding-driven repos to protect the cash lender against counterparty default, and negative in collateral-driven repos to protect the collateral lender against not receiving the collateral back.

Using detailed transaction-level data from 2019 to 2024, we show that, in both funding- and collateral-driven bilateral repos, haircuts are suboptimal from a risk-management perspective. More precisely, we find that haircut setting is shaped by three key frictions. First, dealers' balance sheet constraints play a central role. Regulatory requirements, in particular the Basel III leverage ratio, make it costly for dealers to expand their balance sheets. As a result, dealers tend to set the haircuts in collateral-driven repos higher to limit balance-sheet usage. This effect becomes markedly stronger once the leverage ratio becomes binding in 2021. Second, differences in collateral quality interact with these constraints in an unexpected way. Highly liquid and in-demand government bonds — such as German and French government securities — provide significant non-pecuniary benefits (convenience yield) to market participants. Yet, these high-quality assets often receive higher, not lower, haircuts. This occurs because the dealers most active in these repo market segments are especially balance-sheet-constrained. Third, market power matters. Many repo market participants trade with only one or two dealers, which weakens their ability to negotiate haircuts. As a result, market participants with more dealer connections can negotiate better haircut terms.

These distortions have important consequences for market functioning and monetary policy transmission. When haircuts do not absorb risk, price adjustments take place through more volatile repo rates and trading volumes instead. Moreover, balance-sheet-constrained dealers adjust their repo rates less in response to ECB policy changes, leading to weaker monetary policy pass-through.

A theoretical model supports these findings: higher balance-sheet costs increase haircuts, while the convenience yield of the collateral tends to reduce them. Importantly, the model shows that regulatory constraints exert a stronger influence on haircuts than collateral risk itself.

Overall, the paper highlights how regulation, collateral heterogeneity, and market structure jointly distort haircut setting in euro area repo markets, with implications for market stability and the effectiveness of monetary policy.

1 Introduction

The money market is crucial for the efficient allocation of short-term funding, securities financing and collateral sourcing. Furthermore, a deep and well-functioning money market is a precondition for an efficient monetary policy. Given its essential role in the financial and economic system, it is important to understand the frictions in the money market and how they impede the efficiency of the money markets.

The repurchase agreement (repo) is the main money market instrument, it is a short-term loan secured by different types of government securities. A repo contract is characterized by the (i) volume, (ii) rate, (iii) haircut, and the (iv) a security being posted as collateral. A unique feature of repos is that they can serve a dual role in obtaining either cash or collateral. As a result, repo contracts can either be traded for funding- or for collateral-purposes. If a client counterparty of a dealer intends to obtain funding, repos serve a funding-purpose; if a counterparty is interested in borrowing a specific collateral, they are collateral-driven. Using a unique transaction-level data set of euro area repo transactions for the bilateral dealer to client segment, we want to better understand (i) how different money market frictions impact the haircut setting procedures differently in collateral- and funding-driven repo market environments, and (ii) what are the implications of frictions preventing dealers from setting optimal haircuts.

The basic intuition behind collateralised lending is that haircuts should protect the cash- or collateral-lender from the counterparty's default, and that rates are a function of supply and demand for cash or (sometimes scarce) assets. If a dealer lends out cash (in a *funding-driven* transaction), we would expect haircuts to be positive to protect the dealer against a client default; if a dealer lends out collateral (in a *collateral-driven* transaction), we would expect haircuts to be negative to protect dealers against the risk of the collateral not being returned. Empirically, we, however, find that this is far from being true. For our analysis, we focus on transaction-level data in the bilateral repo segment from the ECB's money market statistical reporting (MMSR) dataset. Overall, haircuts may be set either at the transaction level or at the portfolio level through aggregate margining. To address concerns that portfolio-level margining could distort our transaction-level haircut measures, we show that the distribution of haircuts observed at the transaction level is essentially identical to the distribution implied by dealer-client-day aggregated cash and collateral positions. This close correspondence alleviates concerns that our haircut measures are materially affected by aggregate margining practices. There are several important patterns that we observe in this data, both at transaction-level and at dealer-customer level: First, haircuts are often set at 0%, in which case they are not serving a risk-mitigation purpose, as noted by [Financial Stability Board \(2025\)](#). Second, counter-intuitive

haircuts, such as positive haircuts in collateral-driven transactions and negative haircuts in funding-driven transactions, are common. Third, one would expect the haircut to be collateral-dependent: the scarcer or safer the asset is, the lower the haircut should be in collateral-driven transactions. Our third observation is that this is not the case. Collateral-driven transactions backed by German or French collateral do have higher haircuts, on average, than those backed by Spanish and Italian government bonds. In short, we observe that many repo contracts are priced as if there is no counterparty or collateral re-hypothecation risk, or are priced contrary to ex ante expectations.

In the first part of this paper, we show how three key frictions in money markets impact the pricing of repo contracts. The primary focus of this analysis is to understand how haircuts are determined, if indeed they reflect other considerations than risk management.

We first provide stylised empirical evidence that dealers' balance-sheet capacity puts upward pressure on haircuts in collateral-driven markets but not in funding-driven markets. For this, we add dealer's leverage ratio, which are sourced from the ECB's harmonised financial reporting (FIN-REP) dataset. From a dealer's perspective, collateral-driven repos are balance-sheet costly, as the cash leg adds to their liabilities and counts towards the non-risk-weighted leverage ratio. The lower the haircut, the higher the amount of cash that the dealer will have to take on its balance sheet, which provides an incentive to increase haircuts when balance sheet constraints become more binding. In the *cross-section*, we show that this effect is stronger for dealers that are more balance-sheet constrained (i.e., have a lower leverage ratio). Importantly, dealers in German collateral, for example, are - on average - more balance-sheet constrained than dealers in Italian securities, which provides intuition for why haircuts for collateral countries with higher convenience yields can still be higher. In the *time-series*, we show that this effect has only emerged since the leverage ratio requirements for dealer banks in the euro area have become binding. Importantly, the leverage ratio in the euro area becomes binding only on reporting days (quarter-end days). In line with this, a large part of the literature, such as [Ranaldo et al. \(2021\)](#), have analysed the effect on, for example, rates on those quarter-end days. We show that, in our sample, the effect of the leverage ratio on haircuts is present not only on those days but throughout the entire quarter. Lastly, the leverage ratio regulation in the euro area is asymmetric as it does not penalize reverse repos, hence, there is no effect on funding-driven repos (see Figure 1).

Second, we show that collateral heterogeneity plays an important role not only for rates but also for haircuts. The higher the convenience yield of a collateral asset, the larger the value assigned to it, and thus the risk of losing the collateral. Hence, we observe that haircuts decrease with convenience yields for collateral-driven repo transactions where the dealer fears losing the collateral.

Third, we find that market segmentation impacts haircuts as well. The market is segmented not only in the sense that a few dealers exercise significant market power over their clients, but also in

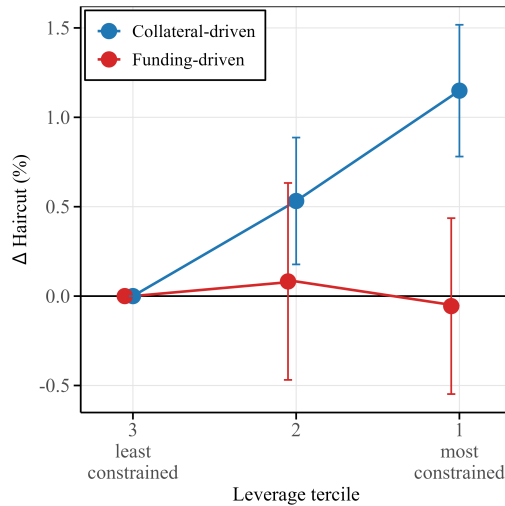


Figure 1: Haircut adjustments to leverage constraints

The plot shows the estimated differences in haircuts relative to the tercile with the least constrained banks (least constrained = 3, most constrained = 1) after the 3% leverage ratio requirement became binding for all EU banks on June 28, 2021. The coefficients and standard errors shown in this graph correspond to regressions results presented in columns (3) and (6) in Table 3. ‘Collateral-driven’ and ‘funding-driven’ defined from the perspective of the dealer’s clients.

that not every dealer or client trades every type of collateral. As a consequence, clients typically trade with just one or two dealers within a specific collateral class, and the contracts for each link tend to be determined more by market convention than by time-varying risks in these markets.

We then propose a theoretical framework that incorporates the effects of market frictions leading to sub-optimally set haircuts into a partial equilibrium model for the repo market. To this end, we are using a Bertrand model with soft as well as strict capacity constraints regarding collateral and cash. The choice of this model is motivated by the fact that the structure of the repo market is highly oligopolistic (Eisenschmidt et al., 2024). With minimal assumptions, we are modelling the collateral- as well as the funding-driven side of the market explicitly and are able to show how the different frictions impact the equilibrium haircut. In particular, the model sensitivities for collateral-driven repos predict that repo haircuts tend to increase in dealer balance sheet cost, while they tend to decrease in the asset convenience yield. Those results are in line with what we observe in the data. Importantly, haircuts are much more sensitive to a change in dealer balance sheet costs than to a change in the asset convenience yield which can explain the puzzle of higher haircuts for collateral-driven repos secured by German as opposed to Italian collateral.

What are the consequences for market stability and monetary policy if the haircut as the key risk management tool of repo contracts is not used as intended? Re-hypothecation risk and default risk are difficult to measure and typically very low in this market. What we can show is that repo rates and repo volumes of transactions with suboptimally chosen haircuts are more volatile. One potential

explanation is that if haircuts are sticky and bound within certain limits, dealers are more likely to adjust their rates and volumes to price in the risks that are not reflected in the repo haircut.

Regarding the transmission to monetary policy, we finally provide evidence that it is affected by dealers' balance-sheet constraints. If, in collateral-driven markets, more constrained dealers choose suboptimal haircuts and instead use repo rates as a risk-mitigation tool, then we expect the monetary policy transmission from dealers to OTC clients to be weaker for more constrained dealers. This is what we find empirically - the more constrained dealers are, the slower do the repo rates that they offer adjust to the new monetary policy rates. These results suggest that the regulatory measures implemented by financial stability authorities have affected monetary policy and may have introduced even more volatility than before.

Our analysis mainly contributes to two strands of the literature. First, we add to the literature on short-term money markets. The innovations we bring are twofold: First, we document how money market friction impacts haircuts being chosen in the repo market. And second, we differentiate those effects for repos that either serve a funding- or collateral-purpose.

The prevalence of zero and negative haircuts is at the centre of a policy debate over the financial stability risks posed by their effects on the leverage of non-bank financial institutions, with one recommendation under discussion being the imposition of minimum haircuts ([Financial Stability Board, 2025](#)). The market practice of zero and negative haircuts is confirmed *inter alia* by [Kahn and McCormick \(2025\)](#) and [Cenicola et al. \(2025\)](#).

Three papers are most closely related to our work. [Julliard et al. \(2019\)](#) investigate the impact of money market frictions on repo haircuts based on a transaction-level dataset that is a snapshot of the repo books of six banks that are major players in the UK repo market. Our work adds several key aspects such as a time series dimension that allows us to, for example, measure the effect of the introduction of the leverage ratio on dealers haircut setting practices, as well as a differentiation between collateral- and funding-driven repo transactions that are differently impacted by money market frictions. [Hermes et al. \(2025\)](#) report that even riskier euro-area collateral frequently carries zero or negative haircuts. We add to this by focusing on dealers' balance sheet constraints and the underlying frictions as well as the implications for financial stability and monetary policy. Focusing on the U.S., [Wallen and Lu \(2025\)](#) document the prevalence of negative haircuts on the U.S. Treasury repo market, and show that major dealer banks sometimes quote negative haircuts on U.S. Treasury repos when they have excess balance sheet capacity, further indicating that haircuts do not solely reflect collateral risk. We contribute to this by examining dealers' regulatory ratios and, in the context of the euro area, the importance of collateral-driven transactions relative to cash-driven transactions in distorting haircuts.

Related to our analysis, studies on safe asset scarcity and collateral “specialness” (e.g., [Arrata et al., 2020](#); [Corradin and Maddaloni, 2020](#)) show that high-quality government bonds in Europe command convenience yields that can impact repo markets. In line with this, [Krishnamurthy and Vissing-Jorgensen \(2012\)](#) illustrate how safe assets carry a premium in investor preferences that translates into repo pricing ([Ballensiefen and Ranaldo, 2023](#)). Focusing more on crisis periods, [Gorton and Metrick \(2012\)](#), for example, documented how repo haircuts spiked during the 2008 financial crisis.

Second, we add to the literature on monetary policy, financial regulation, and the monetary policy transmission process (e.g., [Duffie and Krishnamurthy, 2016](#); [Drechsler et al., 2017](#); [Ballensiefen et al., 2023](#); [Eisenschmidt et al., 2024](#)). It has been shown that post-crisis bank regulations can have unintended effects on short-term funding markets. In particular, [Ranaldo et al. \(2021\)](#) find that the Basel III leverage ratio constraint induces “window dressing” behavior by European banks. We add to this by shedding light on how central bank regulation impacts euro area repo markets throughout the entire quarter, leads to suboptimally chosen haircuts from a risk-management perspective, and thereby adversely impacts the monetary policy transmission. This adds to the work by, for example, [Ballensiefen et al. \(2023\)](#) and [Eisenschmidt et al. \(2024\)](#) on the monetary policy transmission process. We show that haircuts are sub-optimally chosen so that adjustments in central bank rates translate into heightened volatility in repo rates/volumes rather than uniform interest rate pass-through. Such concerns resonate with findings by [Duffie and Krishnamurthy \(2016\)](#) and [Drechsler et al. \(2017\)](#).

Overall, our study contributes to the existing literature by introducing several novel dimensions. First, we specifically incorporate dealers’ balance-sheet constraints by focusing on regulatory ratios, allowing us to directly assess how post-crisis regulation shapes haircut setting. Second, exploiting detailed euro area data, we distinguish between collateral- and cash-driven repo transactions and show that these two segments are affected differently by money market frictions, with important implications beyond haircut formation on repo market outcomes and monetary policy transmission. And third, we add a theoretical model that rationalizes our empirical findings.

The remainder of the paper proceeds as follows: section 2 describes our setting and the data, section 3 documents empirically how frictions in the money market impact repo haircuts in the bilateral repo market segment, section 4 introduces our model, section 5 discusses the consequences of our findings for market stability and the monetary policy transmission, and section 6 concludes.

2 Euro area repo market

The focus of this paper is to provide new insights about how repo markets function in collateral- and funding-driven environments, and to understand what are the implications for financial stability and

the monetary policy transmission process. For this, we focus on the European money market, which provides a unique setting to analyse those questions.

2.1 Euro area repos

In a repo trade, two counterparts exchange cash for collateral for a predefined time period with a fixed repurchase obligation. The lender in a repo transaction provides a short-term loan (over)collateralized by a sovereign bond. Repo transactions are almost exclusively traded with large dealer banks being one counterparty in the repo transaction. The repo market, therefore, breaks down into two broad segments: an *inter-dealer* segment in which dealers trade with one another, and a *dealer-client* segment in which clients such as money market funds, mutual funds, pension funds, or insurance companies trade with dealers. While the inter-dealer market is structured as a centrally cleared market which offers, for example, netting benefits to dealer banks, the dealer-client segment is a bilateral, over-the-counter or OTC market. In our analysis, we focus on the bilateral dealer-client segment of the repo market as we are interested in how money market *frictions* that are prevalent in an OTC market such as dealer balance sheet cost (as there are no netting benefits) or dealer market power shape repo market outcomes.

A repo is legally structured such that the lender benefits from the ability to use the collateral for the time between the purchase and repurchase.¹ Because of the collateralized nature, repos serve a dual role for obtaining cash and collateral. Repos in the bilateral segment are either *funding-driven* (when the client borrows cash from a dealer bank) or *collateral-driven* (when the client borrows a collateral from a dealer).

For our analysis, we use the Money Market Statistical Reporting (MMSR) dataset, collected daily on the basis of transaction-by-transaction data from a list of 53 euro area reporting agents.² The MMSR Regulation requires reporting agents to report their activities in the money markets with all counterparties, with other reporting agents (inter-dealer trades) as well as with non reporting customers (or dealer-client trades). In our analysis, we focus specifically on the segment of dealer-client trades settled bilaterally, as this is where we expect to see frictions at play in haircuts' setting, as in Eisenschmidt et al. (2024). Focusing on this bilateral segment rules out many confounding factors such as homogeneous pricing and haircut policies being automatically applied in the centrally-cleared repo market.

The data set covers the time period from 2019 until June 2024. The secured segment consists of

¹The lender in a repo transaction becomes the legal owner for the term of the repo and thus has a benefit from obtaining a specific bond as collateral (see Duffie (1996)).

²Selected for their importance as dealers in the euro money market, see the list of reporting agents: https://www.ecb.europa.eu/stats/financial_markets_and_interest_rates/money_market/html/index.en.html

all repo transactions denominated in euros with a maturity of up to and including one year³ that can either be conducted via a central counterparty or in bilateral transactions.

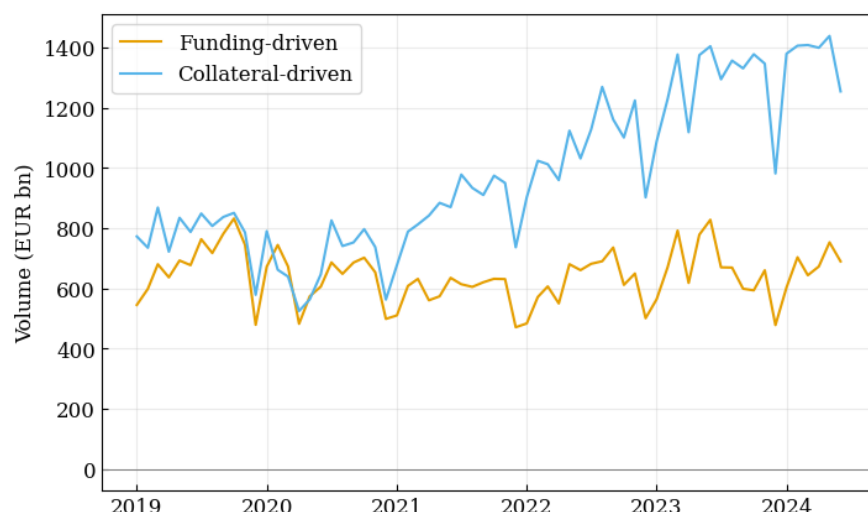


Figure 2: Trading volumes in funding- and collateral-driven repos over time

This graph shows the aggregated monthly volume of one-day repos in the OTC-segment of transactions collateralised by German, French, Italian or Spanish general government bonds.

We classify repo trades into collateral-driven and funding-driven. Our underlying assumption is that repo transactions in the dealer-client segment are client-initiated as dealers could always access the centrally-cleared inter-dealer market which offers several benefits to clearing members such as the aforementioned netting. Hence, we classify repo transactions in which dealers lend out cash as funding-driven (from the perspective of the client) and repo transactions in which dealers lend securities as collateral-driven.⁴ Figure 2 illustrates the aggregated monthly trading volume of funding- and collateral-driven transactions in our data. Overall, the repo market is, on average, more collateral-driven; at the beginning of 2019, volumes in collateral- and funding-driven repo trades are about the same, however, until 2024, volumes have almost doubled for collateral-driven repos, while volumes in funding-driven repos have roughly stayed the same. This is particularly noteworthy since collateral-driven trading has become even more prevalent in a time of falling excess liquidity. Given that a collateral-driven trade can also be seen as a tool to invest cash safely, this underlines the importance of collateral-trade motives. Overall, those developments speak to the importance of the collateral-driven repo market segment and reinforce the importance of understanding how effectively this market works.

³Defined as transactions with a maturity date of not more than 397 days after the settlement date.

⁴An alternative way to classify trades into collateral-driven and funding-driven could be to differentiate between general collateral and special collateral trades. The asset being used as collateral can be a particular asset ('special repo') or any asset from a predefined basket of assets ('general collateral repo'). However, only about 20% of repo transactions in the MMSR data contain a GC vs special flag as reporting this information was not mandatory for reporting dealers in the MMSR data set.

2.2 Repo haircuts

MMSR reporting collects transaction-level haircuts, which in the bilateral repo segment should always reflect a “percentage discount on the initial market value of the collateral”. The haircut is defined as

$$\text{Haircut} = 100 - \left(\frac{\text{Cash lent or borrowed at inception}}{\text{Collateral market value at inception including interest}} \times 100 \right).$$

MMSR reporting guidelines explicitly foresee the case of negative haircuts: “In certain collateral-driven transactions, the value for the field may be negative, representing that the collateral provider requires delivery of a cash amount at inception larger than the market value of the security.”⁵

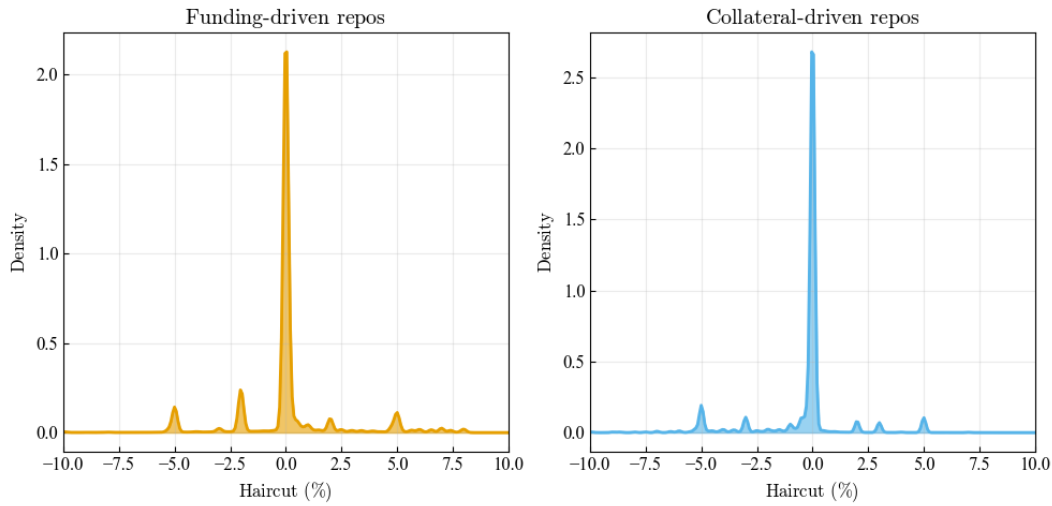


Figure 3: What is the distribution of haircuts?

This histogram displays the estimated distribution of haircuts in funding-driven and collateral-driven repo transactions between dealer banks and non-dealer clients from January 2019 to June 2024. The estimation used a Gaussian kernel.

To recall, for funding-driven repo transactions, haircuts insure the dealer against borrower default, whereas in collateral-driven repos, they protect the dealer against the risk of not receiving back the collateral. As a result, one expects haircuts set by dealers vis-à-vis their clients to be positive in funding-driven transactions, whereas to be negative in collateral-driven transactions. Figure 3 shows in this context the distribution of repo haircuts for funding-driven and collateral-driven repos. The distribution suggest that haircuts are often set counter-intuitively from a risk-management perspective. Two observations are noteworthy: (i) repo haircuts are often times set at 0%, which seems to be a sub-optimal choice from a risk management perspective. (ii) that repo haircuts for funding-driven repos are not strictly positive (or zero), whereas repo haircuts for collateral-driven repos are not strictly negative (or zero), which raises the question of the underlying determinants in the *time-series* and

⁵See Reporting instructions for the electronic transmission of money market statistical reporting (MMSR), version 3.6.1 (30 August 2023) https://www.ecb.europa.eu/stats/money/mmss/shared/files/MMSR-Reporting_instructions.pdf.

cross-sectional haircut variation.

Transaction-level haircut data warrant caution in terms of interpretation. First, it is common practice to *report* zero haircut for cleared transactions, for instance for CCP-cleared repo where the initial margin is the main risk-management tool. In MMSR, as a result, all transactions against a CCP will be reported with a collateral haircut set to zero. In our case, we focus only on the bilateral repo segment, where zero and negative haircuts are well documented across markets, datasets, and jurisdictions (see for instance [Hempel et al., 2023](#); [Kahn and McCormick, 2025](#)). We also check that the distribution of repo haircuts is similar using Securities Financing Transactions (SFT) data, an alternative and more recent dataset available since 2021 (see also [Hermes et al., 2025](#)).

Second, aggregate margining, i.e., the setting of margins at the portfolio-level instead of at the transaction level, may result in aberrant haircuts ([Hill and Westphal, 2025](#)). To address this concern, in the Appendix, we aggregate, at the dealer-client-day level, the cash and the collateral positions to construct an implied aggregate haircut. The resulting haircut distributions for funding- and collateral-driven repos are shown in Figure A.8. The distributions are not materially different from the ones derived from transaction-level data, especially in terms of prevalence of zero and negative haircuts. This mitigates concerns that our results are biased by aggregate margining at the dealer-customer-day level.

Third, negative haircuts may also arise from valuation practices and reflect price variation in the collateral position in repo transactions set with an initial zero haircut, as noted by [Wallen and Lu \(2025\)](#). If haircuts are contractually set to zero by default, our observations would not reflect an active choice by constrained dealers or clients but much rather a technical feature of the repo master agreements. Analysing the within ISIN-dealer-client variation across time, the mean standard deviation in haircuts is 25.5 bps with 17.6% of the ISIN-dealer-clients triplets having a standard deviation in haircuts of more than 10 bps. Comparing across time and clients, we even observe that the mean standard deviation in haircuts within an ISIN-dealer pair is 63.9 bps with 35.6% of the ISIN-dealer pairs having an across client-time variation greater than 10 bps. Altogether, this shows that even for a given relationship and a fixed collateral ISIN, haircuts are actively chosen or at least amended over time.

To better understand the time-series component of repo haircuts, Figure 4 breaks down the repo trading volumes in funding- and collateral-driven repos, separately for transactions where the haircut is set as expected (blue-shaded area) and transactions where the haircut is set sub-optimally (orange-shaded area). What is particularly noteworthy is that the increase in collateral-driven repo trades is almost exclusively driven by repo transactions where the haircut is set sub-optimally, further raising questions as to why this is the case and what are the implications of it.

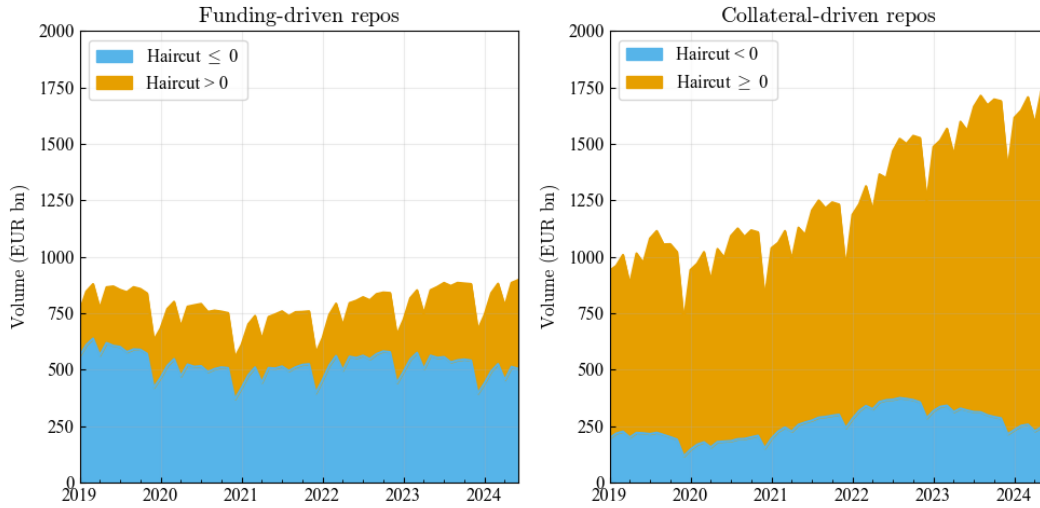


Figure 4: Monthly Aggregate Repo Volume by Haircut Sign

This graph shows the monthly aggregated volume of repos in the OTC market collateralised by German, French, Spanish or Italian government bonds. Additionally, each area is segmented into the volume that stems from transactions having a non-positive, or non-negative haircut, depending on the market. We classify a transaction as funding-driven if the dealer provides cash in the first leg of the repo and receives collateral. Conversely, transactions in which the dealer receives cash and delivers collateral are classified as collateral-driven. Time series leaves out the noise component and only shows the trend and the seasonality component for confidentiality reasons.

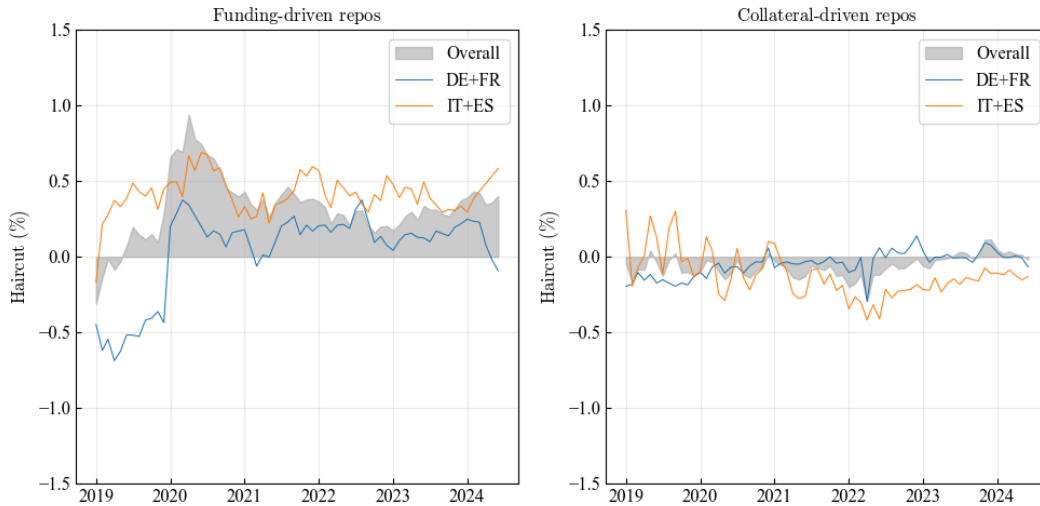


Figure 5: Is there a trend in the average haircut over time?

This chart displays the monthly volume-weighted average haircut in funding-driven repo transactions between dealer banks and non-dealer clients from January 2019 to June 2024. The sample is restricted to bilateral, non-specific collateral trades with overnight maturities (O/N, T/N, S/N), backed by general government securities (S13 sector) issued in Germany (DE), France (FR), Spain (ES), or Italy (IT). Haircuts are bounded within $\pm 100\%$. The dashed line represents the overall average across all included countries.

Figure 5 adds a cross-sectional element to it and plot the time-development of the monthly, volume-weighted average haircut for transactions collateralised by German, French, Italian, or Spanish government bonds, separately for funding-driven and collateral-driven repo transactions. In funding-

Table 1: Volume-weighted Average Haircuts by Collateral Country and Sample Restriction

This table reports volume-weighted average haircuts on one-day repo transactions across four collateral countries (DE, FR, ES, IT). Columns distinguish filtering strategies: (1) no restrictions, (2) restricting to government-sector collateral (S13), and (3) further restricting to dealer–client transactions. Panel A presents collateral-driven repos; Panel B presents funding-driven repos.

Panel A: Collateral-driven Repos				Panel B: Funding-driven Repos			
Country	All	S13 only	S13 + D→C	Country	All	S13 only	S13 + D→C
DE	0.027	-0.012	-0.110	DE	-0.138	-0.157	-0.075
FR	0.331	0.201	0.080	FR	-0.032	-0.069	0.017
ES	0.039	-0.026	-0.186	ES	0.027	-0.019	0.278
IT	-0.046	-0.142	-0.360	IT	0.265	0.183	0.458

driven repos, haircuts across country-collateral are getting more homogeneous since the beginning of 2021. In that period, haircuts are mostly positive and slightly higher for more riskier country collateral such as Spain. As for collateral-driven repos, starting in 2021, dispersion in repo haircuts has increased significantly, with repo haircuts for German and French collateral being higher than repo haircuts for Spanish and Italian collateral. Those results are confirmed by the average haircuts in dealer to client trades for collateral- and funding-driven transactions (column 3 in the below Table 1). On average, haircuts seem to be positive for funding-driven repos, while they are negative for collateral-driven repos.

Lastly, Table 2 provides summary statistics for our main empirical variables. As expected, repo rates and repo haircuts are, on average, lower in collateral-driven repos as opposed to funding-driven repos. By contrast, average trade volumes of 29 respectively 26 million euro as well as average terms of around 5 days are comparable between collateral- and funding-driven repos in the dealer-client segment.

In addition, Panel C in Table 2 adds information on each dealer’s leverage ratio, which are sourced from the ECB’s harmonised financial reporting (FINREP) data set. This ratio is available for each dealer banks at a quarterly level. On average, the dealer leverage ratio is 9.8%, however, there is significant dispersion in how balance sheet constrained dealers are, as some dealers have a leverage ratio of below 3% while others have a leverage ratio above 60%.

As [Schaffner et al. \(2019\)](#) show for the inter-dealer market, there is a significant home bias in dealer trading activity. For example, more than 55% of repo transactions using German collateral involve traders who themselves conduct at least 60% of their repo business in this segment. They argue that the "home bias in bond holdings translates into a preference for conducting repos in the corresponding collateral segment." Taking this into account, we also calculate the average leverage ratio of a dealer borrowing against different country collaterals. The leverage ratios of a dealer borrowing against German and French collateral is thereby lower than the average leverage ratio of a

Table 2: Summary Statistics

The value for the quintiles are reporting the mean values for that variable in the given quintile. We cannot show percentiles due to confidentiality concerns.

Panel A: Collateral-driven Repos							
Variable	mean	std	Q1	Q2	Q3	Q4	Q5
Rate (DFR adjusted, %)	-0.248	0.379	-0.732	-0.249	-0.156	-0.096	-0.007
Haircut (%)	-0.098	1.139	-1.140	-0.019	0.000	0.000	0.672
Volume (EUR bn)	0.029	0.048	0.001	0.003	0.011	0.032	0.099
Maturity (days)	5.529	20.388	1.000	1.000	1.062	2.408	33.137
Panel B: Funding-driven Repos							
Variable	mean	std	Q1	Q2	Q3	Q4	Q5
Rate (DFR adjusted, %)	-0.182	0.524	-0.860	-0.184	-0.068	-0.003	0.204
Haircut (%)	0.522	4.368	-2.055	0.000	0.000	0.137	4.530
Volume (EUR bn)	0.026	0.043	0.001	0.003	0.009	0.027	0.089
Maturity (days)	5.978	26.781	1.000	1.000	1.000	1.931	34.627
Panel C: Dealer characteristics							
Leverage ratio	0.098	0.194	0.026	0.056	0.076	0.101	0.602

dealer borrowing against Italian or Spanish collateral: the leverage ratios are 4.37% for Germany collateral, 4.16% for French collateral, 4.71% for Italian collateral, and 4.92% for Spanish collateral. This implies that (relative to the minimum required leverage ratio of 3%, an average dealer trading in German government bonds is about 18% more capital-constrained than an average dealer in Spanish collateral.

3 Empirical analysis

Intuitively, one would expect haircuts for German or French collateral in collateral-driven transactions to be lower than those haircuts for less thought after collateral such as Italian and Spanish bonds. This raises the question of how frictions in repo markets impact haircut setting practices. The empirical part is, therefore, structured to document how repo-market frictions impact repo-market haircuts. Those empirical results will then be the starting point for developing our repo market model.

In our analysis, we consider three main money market frictions and their impact on repo haircuts: (i) dealer balance sheet constraints, (ii) the convenience yield embedded in the collateral value, and (iii) dealer market power.

3.1 Dealer balance sheet constraints.

The literature so far has shown that dealer regulation in the aftermath of the Global Financial Crisis has impacted dealer intermediation in the repo market. One regulation which has received particular attention is the introduction of the leverage ratio which asymmetrically impacts repos and reverse

repos. If a dealer lends out collateral and receives cash in return (a reverse repo trade from the dealer's perspective), this extends the dealer's balance sheet. By contrast, a repo transaction from the dealer's perspective in lending out cash for collateral does not impact its leverage ratio calculation. [Ranaldo et al. \(2021\)](#) have shown the impact of the leverage ratio regulation on repo rates. In particular, the leverage ratio in the euro area becomes binding on reporting days (quarter-end days). Since it is asymmetric in nature (i.e., entering a repo contract to borrow cash extends a bank's balance sheet, and hence lowers the leverage ratio, whereas entering a reverse repo contract does not), this implies that dealers are more dis-incentivized to lend collateral (collateral-driven repos) than to lend cash (funding-driven repos).

As a first step, we want to understand the impact of the leverage ratio regulation on repo haircuts instead of repo rates on reporting days. Since haircuts for collateral-driven reverse repo trades are negative, these negative haircuts are particularly penalizing the dealer's leverage ratio on reporting days. We, therefore, would expect an upward pressure on repo haircuts for collateral-driven repo trades on reporting days; we do not expect this upward pressure for funding-driven repo trades since they are not negatively impacted by the leverage ratio calculation.

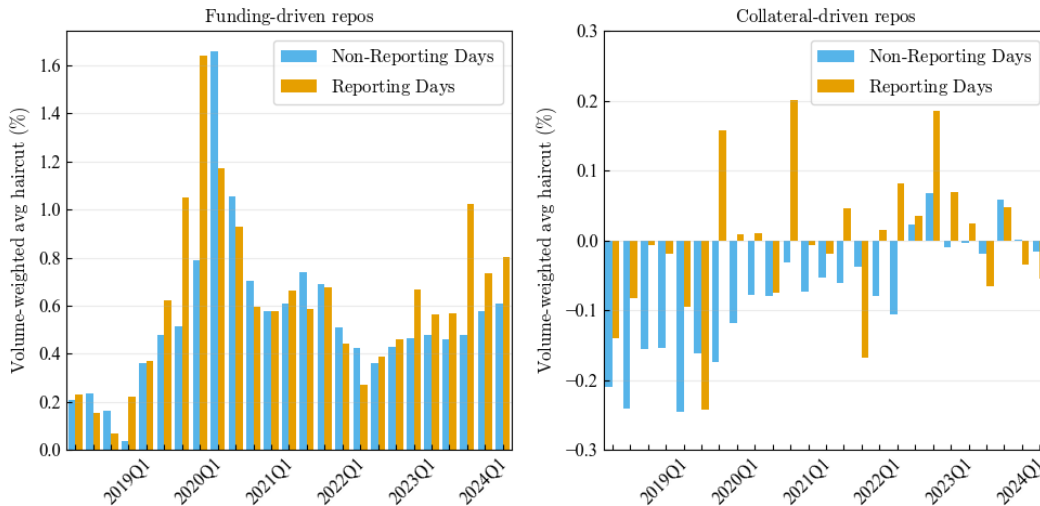


Figure 6: Do haircuts matter for balance sheet space?

The graph plots the volume-weighted average haircut over time of funding- and collateral-driven transactions between the client sector (non-deposit-taking institutions) and the dealer sector. It considers only transactions collateralised by general government bonds. Funding-driven transactions include all non-specific collateral trades, excluding those backed by government bonds from Germany, the Netherlands, Austria, and France. Conversely, collateral-driven transactions encompass trades secured by government bonds from these four countries, while general collateral transactions are excluded. Only transactions with haircuts $\in [-80, 80]$ are considered. Reporting days are identified as the last two days of each quarter, whereas non-reporting days refer to all remaining days of each quarter.

Figure 6 provides a first visual impression of this effect. In general, we observe that average haircuts tend to be positive for funding-driven trades while they are mostly negative for collateral-

Table 3: Do balance sheet constraints matter for haircuts?

The dependent variable is the volume-weighted average haircut. Banks are assigned to leverage terciles each quarter based on their reported leverage ratios, with tercile 1 indicating the most constrained institutions (i.e., lowest leverage ratios). The baseline is the third tercile, i.e., the group of banks with relatively high leverage ratios (and low balance sheet constraints). The variable *Post* equals one for transactions occurring after June 2021, and zero otherwise. The interaction terms capture the change in haircut behaviour for each leverage-constrained group following this date. Columns (1)–(3) report results for collateral-driven markets; columns (4)–(6) repeat the specifications using funding-driven markets. The sample includes repo transactions between January 2019 and June 2024, restricted to short maturities (O/N, T/N, S/N, 1W), where either the borrower (columns 1–3) or the lender (columns 4–6) is an ECB-identified dealer, and the counterparty is not. Haircuts are limited to transactions with valid collateral characteristics and where the collateral is issued by general government institutions (S13 sector). Standard errors clustered at the dealer-level are reported in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

	Haircut (%)					
	Collateral driven			Funding driven		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Tercile1</i> $\times$ <i>Post</i>	1.186*** (0.209)	1.233*** (0.206)	1.149*** (0.188)	−0.097 (0.657)	0.261 (0.510)	−0.056 (0.251)
<i>Tercile2</i> $\times$ <i>Post</i>	0.652*** (0.131)	0.704*** (0.162)	0.532** (0.181)	0.016 (0.487)	0.164 (0.430)	0.082 (0.281)
<i>N</i>	64 064	64 064	64 064	39 940	39 940	39 940
<i>R</i> ²	0.22	0.49	0.60	0.18	0.43	0.61
FE: Coll. country	X	X	X	X	X	X
FE: Coll. Sector	X	X	X	X	X	X
FE: Client		X			X	
FE: Dealer $\times$ Client			X			X
FE: Month	X	X	X	X	X	X
FE: Leverage Tercile	X	X	X	X	X	X

driven repo trades. However, while repo haircuts for funding-driven repo trades do not seem to differ between reporting and non-reporting days, we do observe a clear upward pressure on repo haircuts on reporting days, which goes hand-in-hand with the negative repo rate impacts on reporting days as has been documented in, for example, [Ranaldo et al. \(2021\)](#).

However, challenging the conventional wisdom, we go one step further and want to understand whether repo haircuts over the entire quarter are driven by the leverage ratio. The rationale behind it would be that dealers do not only adjust repo haircuts on reporting days, but more generally set repo haircuts over a given quarter based on their capital constraints. This means we want to better understand the *cross-sectional* variation in dealer balance sheet constraints as well as their implications on repo haircut practices. To capture this effect, we sort dealers in each quarter into different leverage ratio terciles. Dealers in tercile 1 are the group of banks with the lowest leverage ratios (i.e., most constrained dealers), whereas the group of dealers in tercile 3 are the ones which are least balance sheet constrained. Importantly, we introduce the variable *Post* equal to one occurring for transactions after June 2021 when a binding 3% leverage ratio requirement was implemented for all banks supervised by the European Central Bank (ECB),

Table 3 provides evidence for this intuition. More precisely, we analyse how the haircuts of the more constrained dealers in terciles one and two change (relative to the more unconstrained dealers in tercile 3) in the period after the leverage ratio requirements became binding, i.e., after June 2021. The results provide clear evidence that repo haircuts in collateral-driven repo transactions significantly increase in the leverage ratio constraints of a given dealer, and in the period after the introduction of the leverage ratio requirements. The monotonic pattern provides evidence that more constrained dealers, indeed, tend to charge higher haircuts, providing a first explanation for why haircuts in collateral-driven repo transactions are often observed to be zero or positive (instead of being negative). This effect, as expected, is not present in funding-driven repo transactions. The results are present for different fixed effect specifications such as collateral country, client, dealer-client, time, or leverage tercile. Reported standard errors are clustered at the dealer-level.

Overall, the results on dealer balance sheet constraints provide several key insights into how repo haircuts are being determined. First, in the cross-section, we show that repo haircuts, in addition to repo rates, are significantly impacted by the Basel III leverage ratio regulations. In particular, due to the asymmetric treatment of repo and reverse repo trades, dealers, in particular those that are more balance sheet constrained, have an incentive to charge non-negative haircuts in order to reduce their leverage. This prevents haircuts from being optimally set in collateral-driven repo transactions. Second, in the time-series, we show that this effect is not only prevalent on reporting days itself, but over the entire quarter, indicating that the euro-area quarter-end regulation requirements not only impact repo markets on quarter-end trading days.

3.2 Convenience yield

Investors pay a “convenience premium” for the safety and liquidity benefits provided by safe assets (Gorton, 2017) which is also reflected in repo rates (Ballensiefen and Ranaldo, 2023). Prior research has shown that the convenience premium varies across securities (Krishnamurthy and Vissing-Jorgensen, 2012) and market conditions, such as the opportunity cost of holding money (Nagel, 2016) and uncertainty (Moreira and Savov, 2017). However, our understanding of the haircut implications of the temporal and cross-sectional variation in convenience premiums is still limited.

Table 4 is a first approach to capture the effect of convenience yield on repo market haircuts. The convenience yield is measured as the difference between the ECB’s DFR and the average repo rate for a given country in a given month. Across various fixed effect specifications, we observe that an increasing specialness is negatively related to repo haircuts. This effect is present in collateral-driven markets where the dealer fears to lose the collateral asset and the future convenience yields he has assigned to it.

Table 4: Convenience yields and haircuts

The dependent variable is the volume-weighted average haircut. The convenience yield is calculated as the spread between the ECB's DFR and the average repo rate for a given country in a given month. Columns (1)–(3) report results for collateral-driven markets; columns (4)–(6) repeat the specifications using funding-driven markets. The sample includes repo transactions between January 2019 and June 2024 where either the borrower (columns 1–3) or the lender (columns 4–6) is an ECB-identified dealer, and the counterparty is not. Standard errors clustered at the dealer-level are reported in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

	Haircut (%)					
	Collateral-driven			Funding-driven		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>ConvenienceYield</i>	−0.706*** (0.101)	−0.689*** (0.087)	−0.606*** (0.066)	0.122 (0.166)	0.195 (0.223)	0.469 (0.355)
<i>Volume</i>	0.260 (0.370)	−0.261 (0.206)	0.147 (0.190)	−0.433 (0.765)	−1.325*** (0.419)	−1.161*** (0.316)
<i>MaturityBucket</i>	−0.003*** (0.001)	0.001** (0.000)	0.001*** (0.000)	0.000 (0.001)	−0.002 (0.002)	0.001 (0.001)
<i>ES</i>	−0.173*** (0.051)	−0.028 (0.040)	−0.087** (0.035)	0.185** (0.093)	0.195*** (0.057)	0.182*** (0.059)
<i>FR</i>	0.033 (0.034)	0.010 (0.022)	−0.046*** (0.014)	0.153 (0.240)	0.110 (0.096)	0.043 (0.046)
<i>IT</i>	−0.267*** (0.053)	−0.071** (0.035)	−0.122*** (0.033)	0.426*** (0.092)	0.274*** (0.072)	0.207*** (0.077)
<i>N</i>	37 589	37 589	37 589	25 915	25 915	25 915
<i>R</i> ²	0.04	0.39	0.56	0.00	0.35	0.68
FE: Month	X	X	X	X	X	X
FE: Client		X			X	
FE: Client × Dealer			X			X

Table 5: Does dealer market power matter for haircuts?

The dependent variable is the transaction haircut. Each regression on the left hand side is based on a subset of all *repo* transactions conducted by MMSR-reporting banks, specifically focusing on bilateral, non-generic collateral transactions secured by general government bonds. Each regression on the right hand side is based on a subset of all *reverse repo* transactions conducted by MMSR-reporting banks, specifically focusing on bilateral, non-specific collateral transactions secured by general government bonds. Only transactions with haircuts within the range $[-100, 100]$ are considered. Standard errors are clustered at the dealer-client-level and reported in parentheses. $*p < 0.1$; $**p < 0.05$; $***p < 0.01$.

	Collateral-driven			Funding-driven		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>#Dealers</i>	−0.006 (0.008)	−0.006 (0.008)	0.001 (0.008)	−0.105*** (0.024)	−0.105*** (0.024)	−0.099*** (0.023)
<i>Volume</i>		−0.187 (0.408)	−0.111 (0.391)		1.413 (1.759)	−0.177 (1.642)
<i>Maturity</i>		0.015 (0.010)	0.002 (0.010)		0.018 (0.029)	−0.060 (0.046)
<i>N</i>	1 804 329	1 804 329	1 804 329	1 361 820	1 361 820	1 361 820
<i>R</i> ²	0.13	0.13	0.18	0.24	0.24	0.28
FE: Month	X	X	X	X	X	X
FE: Coll. country	X	X	X	X	X	X
FE: Coll. sector	X	X	X	X	X	X
FE: Dealer			X			X

3.3 Dealer market power and market conventions

A third key friction in bilateral money market segments reflects dealer market power. Because establishing trading relationships in an OTC market is costly and time consuming, market participants are only able to form a few links with dealers, implying that dealer banks have market power over their OTC customers [Eisenschmidt et al. \(2024\)](#).

To analyse the impact of dealer market power on repo haircuts, we calculate the number of dealers a customer has traded with over the previous twelve months. We implicitly assume that clients with more *active* dealer connections have more negotiation power (and dealers have less market power) in adjusting repo haircuts. Clients in the collateral-driven repo market segment would be incentivized to adjust repo haircuts upwards (i.e., lending less cash for a given amount of securities received), whereas clients in the funding-driven repo market segment would have the opposite incentive in negotiating repo haircuts downwards (i.e., borrowing more cash for a given amount of securities posted as collateral).

We formalize this intuition in a set of panel regressions. Table 5 reports the results of the time-series regression: the dependent variable is the transaction haircut, while *#Dealers* represents the number of dealers a customer has traded with over the previous twelve months. We also control for repo volume and the repo term. The results speak to the idea that clients with more dealer relationships tend to obtain lower haircuts in the funding-driven segment. At the same time, we do not observe a significant impact of weaker dealer market power on haircuts in collateral-driven seg-

ments, potentially driven by the fact that the variation in the $\#Dealers$ variable is quite limited for collateral-driven repos (more so than for funding-driven repos).

3.4 Takeaways

The implications of the empirical results are quite notable: In collateral-driven repos, we observe upward pressure on haircuts due to dealer balance sheet constraints, while asset convenience yield exerts downside pressure, leaving the door open for repo rates secured by government bonds that carry a high convenience yield to have relatively higher haircuts as long as the dealers being active in this market are relatively more balance sheet constrained (which is the case, at least, on average). For funding-driven repos, there are no dealer balance sheet constraints, yielding a more monotonous pattern of higher (lower) haircuts for repos secured by government bonds with lower (higher) convenience yield.

4 Model

The idea of the model is to reflect those empirical facts in a theoretical setting.

4.1 Collateral-driven repo market

We consider a stylized model of an oligopolistic repo market with two dealers (indexed by $i \in \{1, 2\}$) who compete to offer contracts to a representative client. The client receives collateral C_i in exchange for providing cash V_i . At the second leg of the transaction, the dealer repays the cash at a repo rate r_i while the client returns the collateral.

The dealers enter a Bertrand competition with collateral pre-commitments in the style of [Kreps and Scheinkman \(1983\)](#). In the first step of the game, each dealer pre-commits to a certain quantity of collateral that it can lend out to the market. In a second step dealers compete in the repo market by offering a competitive contract (r, h, C) . In this way, each dealer faces a strict capacity constraint $C_i^{\max}$ regarding the collateral they can post in step 2 (as there is limited supply of scarce collateral in this market limiting the dealer's inventory), and faces a soft capacity constraint regarding its balance sheet (of the form of increasing marginal costs for absorbing cash V_i onto their balance sheet $\Phi_i(V_i)$). Clients are price-taking agents who choose the dealer offering the lowest effective cost of trade, denoted by $\rho_i = -r_i - \lambda h_i$, where r_i is the repo rate and h_i is the haircut.

In the second stage, given their pre-committed collateral amount C_i , each dealer offers a pair

(r_i, h_i) , where r_i is the repo rate and h_i is the haircut. The dealer receives cash

$$V_i = (1 - h_i)C_i \quad (1)$$

The client chooses the dealer i that offers the lowest effective cost:

$$\rho_i = -r_i - \lambda h_i \quad (2)$$

Dealer i 's expected profit is given by

$$\pi_i = \left[r_{\text{DFR}} - (1 - p)r_i - \frac{y + pL}{1 - h_i} \right] (1 - h_i)C_i - \Phi_i((1 - h_i)C_i) \quad (3)$$

where r_{DFR} is the interest the dealer earns on cash deposited at the central bank, p is the probability of borrower default (or rehypothecation risk), L is the loss given default (as a share of collateral) and y is the convenience yield of holding the collateral. The continuous function $\Phi_i(V)$ represents the total balance sheet cost function for dealer i , defined as:

$$\Phi_i(V) = \int_0^V \phi_i(s) ds \quad (4)$$

where $\phi_i(s)$ is increasing in s . Specifically, we will assume that each dealer receives cash $V = (1 - h)C$ in exchange for posting collateral C , and incurs a quadratic balance sheet cost

$$\Phi(V) = \frac{\gamma}{2} V^2 = \frac{\gamma}{2} (1 - h)^2 C^2. \quad (5)$$

Dealers first simultaneously choose C_i anticipating the other dealer's supply and optimised dealer profit given their internal balance sheet costs for intermediating the collateral demand. In step 2, dealers optimise (r_i, h_i) to maximize π_i under the constraint that the borrower chooses the lowest ρ_i . Each dealer i chooses (r_i, h_i) to maximize expected profit, subject to the collateral capacity constraint

$$\max_{(r_i, h_i)} \pi_i = \left[r_{\text{DFR}} - (1 - p)r_i - \frac{y + pL}{1 - h_i} \right] (1 - h_i)C_i - \Phi_i((1 - h_i)C_i) \quad (6)$$

$$\text{subject to } C_i \leq C_i^{\max} \quad (7)$$

The price for each contract is comprised of the rate and the haircut. We will assume that the client's preferences over these two parameters is given by

$$\rho_i = -r_i - \lambda h_i. \quad (8)$$

The client's demand function is given by

$$D(\rho) = \alpha - \beta\rho. \quad (9)$$

If both dealers offer the same cost ρ to their clients, we will assume that the demand is split evenly across dealers, i.e., each dealer gets 50% of the market share. The equilibrium is a set of two contracts $\{Q_i^* \equiv (C_i^*, r_i^*, h_i^*) \in \mathbb{R}^{>0} \times \mathbb{R} \times (-1, 1) \mid i \in \{1, 2\}\}$ if

$$\pi_i(Q_i^*, Q_j^*) \geq \pi_i(Q_i, Q_j^*), \quad Q_i^* \neq Q_i, \quad \forall i \in \{1, 2\}. \quad (10)$$

Equilibrium in a symmetric market

We will first solve the problem with the simplifying assumption that the dealers are completely symmetrical and a market clearing condition which is customary in these games.

Assumption 1. *The market clears at equilibrium prices, or more formally*

$$C_1^* + C_2^* = D(\rho^*). \quad (11)$$

One would intuitively expect that, in equilibrium, each dealer offers the same contract. The following proposition shows that this is indeed the case.

Proposition 1. *In a symmetric market, there is one unique pure strategy Nash equilibrium in which both dealers offer the same contract (C^*, r^*, h^*) , which is determined by solution to 6 with respect to C , given the optimal rate $r^*(C^*)$ and haircut $h^*(C^*)$. If in the first stage dealer i names anything other than the Cournot quantity C^* , dealer j always sets the price according to its break-even condition.*

Proof. Existence: Clearly, it would not make sense for the dealer to deviate in the first step as any deviation will result in zero profits. If it deviates in the second step, then changing r or h in a way that lowers ρ would give this dealer more demand but he cannot sell more than the pre-specified C^* .

Uniqueness: [Kreps and Scheinkman \(1983, Proposition 2\)](#). One might wonder whether there are several optimal choices for (r^*, h^*) given ρ . It is easy to verify that given ρ , the utility function π_i has a unique maximum on $(r, h) \in \mathbb{R} \times (-1, 1)$. Hence, for each client cost ρ we can assume that in equilibrium, there is one uniquely determined tuple (r, h) corresponding to it. $\square$

Assuming that the equilibrium allocations are exactly symmetric allows us to calculate the optimal haircut and rate explicitly. We will also assume that the parameter γ is such that providing collateral C^* can still be profitable for the dealer. Hence, each dealer has an incentive to provide all collateral

so that in equilibrium, demand at ρ^* is $2C^* > 0$. For ease of notation, we will write $C \equiv C^*$ in the following. We will solve the problem recursively by first assuming that C is fixed in the second part of the game for both dealers, which allows us to compute the optimal tuple (r^*, h^*) . Once we have an expression for r and h in dependence of C , we can then solve for the optimal collateral amount C in the first step of the game. The client chooses the dealer with the lowest effective cost

$$\rho = -r - \lambda h \quad \Rightarrow \quad r = -\rho - \lambda h. \quad (12)$$

The client's demand function is given by

$$D(\rho) = \alpha - \beta \rho. \quad (13)$$

Hence, the equilibrium rate only depends on the haircut

$$r^* = \frac{2C - \alpha}{\beta} - \lambda h^*. \quad (14)$$

The dealer's expected profit function is

$$\pi(r, h) = \left[r_{\text{DFR}} - (1-p)r - \frac{y + pL}{1-h} \right] (1-h)C - \Phi((1-h)C) \quad (15)$$

Substituting equation 14 gives

$$\pi(h^*) = \left[r_{\text{DFR}} - \frac{(1-p)(2C - \alpha)}{\beta} + (1-p)\lambda h^* - \frac{y + pL}{1-h^*} \right] (1-h^*)C - \Phi((1-h^*)C) \quad (16)$$

The first order derivative is given by

$$\frac{d\pi}{dh} = C \left[-r_{\text{DFR}} + \frac{(1-p)(2C - \alpha)}{\beta} + (1-p)\lambda - 2(1-p)\lambda h \right] - \frac{d\Phi(V)}{dh}. \quad (17)$$

Equalising to zero and collecting terms we get the equilibrium haircut

$$h^* = \frac{(1-p)\left(\frac{2C - \alpha}{\beta} + \lambda\right) - r_{\text{DFR}} + \gamma C}{2(1-p)\lambda + \gamma C}. \quad (18)$$

From equation 14 it directly follows that

$$r^* = \frac{2C - \alpha}{\beta} - \lambda \frac{(1-p)\left(\frac{2C - \alpha}{\beta} + \lambda\right) - r_{\text{DFR}} + \gamma C}{2(1-p)\lambda + \gamma C}. \quad (19)$$

4.2 Funding-driven markets

The model of the funding-driven repo market looks very similar to the collateral-driven model with the crucial difference that balance sheet costs are absent as the leverage ratio regulation is asymmetric and lending cash for collateral does not extend the dealer's balance sheet. Again, we have two dealers (indexed by $i \in \{1, 2\}$) who compete to offer cash to a representative client against collateral. The client seeks to borrow cash V_i by posting collateral C_i . At the second leg of the transaction, the client repays the cash at a repo rate r_i , and the dealer returns the collateral.

Each dealer faces a strict capacity constraint $V_i \leq V^{\max}$ on the amount of cash they can lend (e.g., driven by liquidity or capital limits). The client is modelled as a price-taking agent who chooses the dealer offering the lowest effective borrowing cost, denoted by $\rho_i = r_i + \lambda h_i$, where r_i is the repo rate and h_i is the haircut. The parameter $\lambda > 0$ reflects the client's collateral sensitivity.

Each dealer offers a pair (V_i, r_i, h_i) where r_i is the repo rate, h_i is the haircut, and V_i is the loan volume. Given haircut h_i , the amount of collateral posted is:

$$C_i = \frac{V_i}{1 - h_i} \quad (20)$$

Dealer i earns profit from lending V_i at rate r_i , adjusted for risk and collateral value:

$$\pi_i = \left[(1 - p)r_i - pL + \frac{y}{1 - h_i} \right] V_i, \quad (21)$$

Where p is the probability of default, L is the loss given default (per unit of collateral) and y is the convenience yield from holding the collateral.

Equilibrium

The way that this model is set up, it is essentially a Bertrand model with two oligopolists and strict capacity constraints. In the standard economic setting, there is one unique equilibrium, which yields the Cournot outcome. The next proposition shows that the same arguments can be transferred to our setting.

Proposition 2. *The funding-driven market has one unique Nash equilibrium in which both dealers supply their full capacity V_i on the market and charge the same price (r^*, h^*) which optimises the dealer's profit given ρ^* , where ρ^* is determined by $D(\rho^*) = V_1 + V_2$.*

In equilibrium, both dealers offer their full capacity, $V_i = V_i^{\max}$, and compete by minimizing the effective borrowing cost $\rho_i = r_i + \lambda h_i$. In a Bertrand setting with capacity constraints, the equilibrium

price ρ^* is determined by the market-clearing condition:

$$D(\rho^*) = V_1^{\max} + V_2^{\max}. \quad (22)$$

Assuming a linear demand function:

$$D(\rho) = a - b\rho, \quad (23)$$

the equilibrium effective borrowing cost is:

$$\rho^* = \frac{a - (V_1^{\max} + V_2^{\max})}{b}. \quad (24)$$

Each dealer maximizes profit:

$$\pi_i = \left[(1-p)r_i - pL + \frac{y}{1-h_i} \right] V_i^{\max}, \quad (25)$$

subject to the pricing constraint:

$$r_i + \lambda h_i = \rho^*. \quad (26)$$

Substituting the constraint into the profit function and optimizing with respect to h_i yields the optimal haircut:

$$h_i^* = 1 - \sqrt{\frac{y}{(1-p)\lambda}}. \quad (27)$$

The corresponding optimal repo rate is:

$$r_i^* = \rho^* - \lambda + \sqrt{y(1-p)\lambda}. \quad (28)$$

In equilibrium, both dealers offer the same contract (r^*, h^*) , and total market demand equals total capacity. The solution directly depends on the demand parameters a and b , as well as the total capacity supplied.

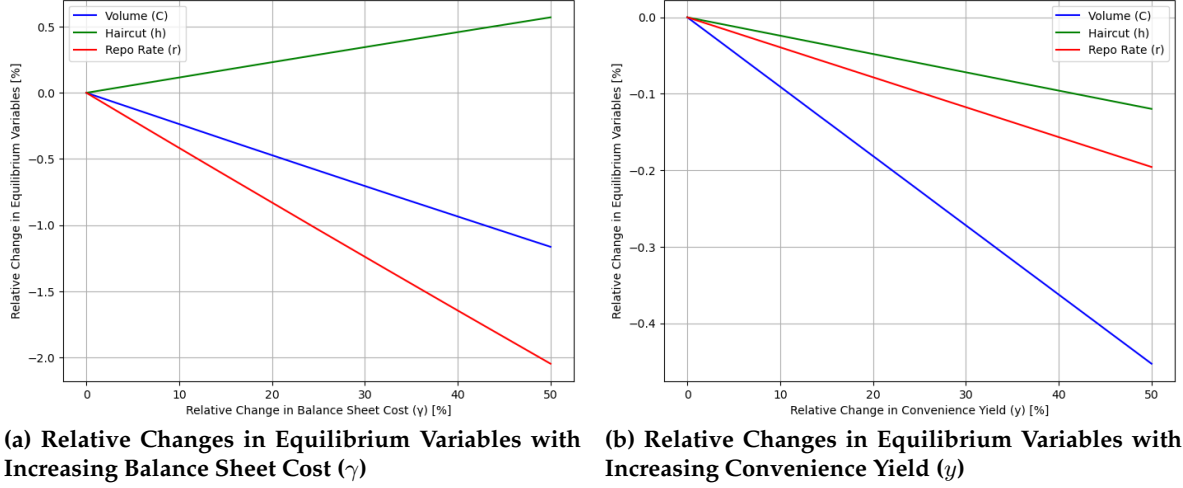
4.3 Comparative statics

Roughly matching the model to the data allows us to analyse how the jointly determined equilibrium tuple (C, h, r) reacts to increased convenience yields or balance sheet costs. Rather than giving us

precise quantitative estimates, this simple model serves first to rationalise the market dynamics that we observed in the empirical parts. The charts presented in Figure 7 show the percentage change in the three outcome variables of a repo contract in response to an $x\%$ change in the balance sheet cost γ , or the convenience yield y .

Figure 7: Sensitivity analysis

These graphs show the relative change of equilibrium contracts in response to a change in the balance sheets costs or the convenience yield. The model was calibrated to match the average transaction backed by German collateral.



The equilibrium dynamics align with three key empirical observations. First, increasing balance sheet costs lead to higher equilibrium haircuts, reinforcing the stylised facts presented in Table 3. Consistent with empirical evidence, although not the primary focus of this paper, higher balance sheet costs are also associated with lower equilibrium repo rates, as dealers require greater compensation for absorbing cash onto their balance sheets. Second, rising convenience yields result in lower haircuts, repo rates, and volumes, in line with the patterns observed in Table 4. Third, the sensitivity of haircuts to balance sheet costs is stronger than their sensitivity to convenience yields. Combined with the empirical fact that dealers intermediating German or French government bonds have had lower average leverage ratios than those intermediating Spanish or Italian government bonds, this helps explain why average haircuts in collateral-driven markets tend to be higher for German and French collateral than for Spanish and Italian collateral.

5 Implications for market stability and monetary policy

So far, we have provided empirical evidence and a theoretical rationale for how key money market frictions impact the pricing of repo contracts in this market. In this section, we are going a step further and ask whether these frictions affect the functioning of this market and the broader transmission of

monetary policy.

5.1 Market stability

If the haircut, as one essential pricing mechanism in repo contracts, is used suboptimally, does this have implications for market stability? This is difficult to answer, as defaults on both the collateral and cash sides of repo transactions are rare, though not entirely absent. What we can examine more thoroughly is whether the repo market became more volatile in terms of rates and volumes in contracts where the haircut was chosen suboptimally. Unlike equity markets, where volatility is often seen as an opportunity for traders, participants in fixed income markets generally prefer stable collateral supply and stable funding conditions. Our idea is that, because the haircut is no longer actively used as the primary measure to manage risk, the rate and volume become more volatile, as they take on a more essential role in controlling the various market risks.

To test this, we first pool repo transactions into negative and non-negative haircuts for collateral-driven transactions and positive and non-positive transactions for funding-driven transactions and then calculate the standard deviation for each month-dealer-country combination in each haircut group.

We then estimate for collateral-driven repo transactions

$$StdRate_{d,t,i} = \beta_1 Haircut_{d,t,i}^{\geq 0} + \gamma_i + \delta_t + \omega_d + Volume_{d,i,t} \quad (29)$$

and

$$StdVolume_{d,t,i} = \beta_1 Haircut_{d,t,i}^{\leq 0} + \gamma_i + \delta_t + \omega_d + Rate_{d,i,t}, \quad (30)$$

where $Haircut_{d,t,i}^{\leq 0}$ is a dummy variable that takes the value one for each given dealer-month-collateral country level that had non-negative haircuts. We also include collateral country fixed effects γ_i , month fixed effects δ_t , and dealer fixed effects ω_d . The variables $Volume_{d,i,t}$ and $Rate_{d,i,t}$ are the averages on the dealer, country and month level. To gauge the effect for funding-driven markets, we define the $Haircut_{d,t,i}^{\geq 0}$ is a dummy variable that takes the value one for each given dealer-month-collateral country level that had non-positive haircuts.

The results are reported in Table 6. Across all specifications and both markets, the coefficient on $Haircut_{d,t,i}$ is positive and significant, meaning that repo contracts with suboptimally chosen haircuts are correlated with higher volatility in rates and volumes.

Panel A focuses on collateral-driven transactions and documents that transactions with 0% and

Table 6: If haircuts are zero, does the risk spill over to the rate and volume?

The dependent variable is the monthly standard deviation of rates (Panel A), and volumes (Panel B) on a dealer, month, maturity, and collateral-country level for transactions with non-zero and zero haircuts separately. The variable *ZeroHaircut* is a dummy taking the value 1 if the average was calculated with the subset of all zero-haircuts in that month-dealer-collateral-country triple, and 0 otherwise. Each regression is based on a subset of all reverse repo transactions conducted by MMSR-reporting banks, specifically focusing on bilateral, non-generic collateral transactions secured by general government bonds. Standard errors are clustered at the dealer-month-level and reported in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Panel A: Collateral-driven								
	Rate (std. dev.)				Volume (std.dev.)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Haircut</i> ≥ 0	0.018*** (0.004)	0.033*** (0.004)	0.039*** (0.004)	0.037*** (0.004)	0.004*** (0.001)	0.006*** (0.001)	0.006*** (0.001)	0.004*** (0.001)
<i>Volume</i>				0.001*** (0.000)				0.001*** (0.000)
<i>N</i>	11 300	11 300	11 300	11 300	11 300	11 300	11 300	11 300
<i>R</i> ²	0.08	0.12	0.22	0.22	0.30	0.36	0.37	0.41
FE: Coll. country	X	X	X	X	X	X	X	X
FE: Dealer		X	X	X		X	X	X
FE: Month			X	X			X	X

Panel B: Funding-driven								
	Rate (std. dev.)				Volume (std.dev.)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Haircut</i> ≤ 0	0.062*** (0.004)	0.075*** (0.004)	0.077*** (0.005)	0.076*** (0.005)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.001*** (0.000)
<i>Volume</i>				0.002*** (0.000)				0.001*** (0.000)
<i>N</i>	12 916	12 916	12 916	12 916	12 916	12 916	12 916	12 916
<i>R</i> ²	0.07	0.09	0.14	0.14	0.32	0.36	0.37	0.42
FE: Coll. country	X	X	X	X	X	X	X	X
FE: Dealer		X	X	X		X	X	X
FE: Month			X	X			X	X

positive haircuts exhibit a significantly higher standard deviation in repo rates compared to negative repo haircut transactions, pointing towards the idea that in those transactions, haircuts are not used as a risk mitigation tool to protect the dealer against the collateral re-hypothecation risk, so that risk not captured in the haircut transmits into the rates leading to a higher repo rate variability. In line with the findings for repo rates, we also observe higher volatility in repo volumes.

Panel B extends the analysis to the funding-driven repo market. Here, we would expect dealers to charge positive haircuts in order to protect themselves against the risk of a borrower default. In line with the results in the collateral-driven segment, we again observe that repo transactions with negative haircuts exhibit a higher repo rate volatility. As the haircut is not used to protect the dealer against the borrower default risk, that risk again transmits into rates leader to higher repo rate volatility.

5.2 Monetary policy transmission

Our results also have *suggestive* implications for the monetary policy transmission process. The “monetary policy transmission pipeline” involves three distinct steps: (i) the central bank’s monetary policy actions pass-through into the money market, (ii) the conditions in the money market impact debt and equity markets, and (iii) monetary policy changes then propagate into the real sector. Focusing on the first step, “in an idealized money market, any change in the main monetary policy rate should pass through perfectly to all money market rates” (Corradin et al., 2020, p.13).

To measure the monetary-policy pass-through, we consider the ten increases in the ECB’s deposit facility rate or DFR that occurred between between July 2022 and September 2023. For each event t and each dealer d , client c , and collateral country i combination, we are first computing the value-weighted average pre and post-announcement rates. The pass-through is then calculated as

$$Passthrough_{d,c,i,t} = \frac{Rate_{d,c,i,t}^{post} - Rate_{d,c,i,t}^{pre}}{DFR_t^{post} - DFR_t^{pre}}. \quad (31)$$

To additionally control for collateral ISIN and maturity-specific fixed effects, we consider the repo rate residuals after regression on collateral ISIN and maturity bucket. The intuition of this measure is that a value close to one indicates an efficient pass-through as repo rates adjust in accordance with changes in the deposit facility rate. A lower pass-through measure, on the contrary, indicates that repo rates do not react in a one-to-one fashion to changes in the DFR.

Returning to dealer balance sheet constraints as a key money market friction, Ranaldo et al. (2021) shows that repo rates at quarter ends exhibit massive spikes, indicating that the repo rate is heavily affected by balance sheet constraints. If, in collateral-driven markets, more constrained dealers

Table 7: Is the transmission of monetary policy affected?

This table shows how the passthrough to clients varies with the leverage ratio of their dealer. FR, IT, and ES are dummy variables equal to one if the repo involves French, Italian, and Spanish government collateral, respectively. We control for the average maturity and haircut of repos between each dealer-customer pair. We also include client and time fixed effects and cluster standard errors by dealer. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

	Passthrough			
	(1)	(2)	(3)	(4)
<i>LeverageRatio</i>	0.831** (0.338)	0.860** (0.374)	0.940*** (0.342)	0.869** (0.385)
<i>ES</i>	0.056* (0.029)	0.059** (0.028)	0.058** (0.028)	0.064** (0.027)
<i>FR</i>	0.016 (0.025)	0.017 (0.025)	0.026 (0.025)	0.032 (0.025)
<i>IT</i>	0.019 (0.023)	0.021 (0.021)	0.045* (0.026)	0.052** (0.022)
<i>Volume</i>			−0.018 (0.275)	0.033 (0.230)
<i>Maturity</i>			−0.006*** (0.002)	−0.006*** (0.002)
Client	Yes	Yes	Yes	Yes
Gov. Council	No	Yes	No	Yes
Num.Obs.	3231	3231	2966	2966
R2 Adj.	0.151	0.485	0.150	0.502

choose haircuts suboptimally and instead use repo rates as a risk mitigation tool, then we expect the monetary policy transmission from dealers to OTC clients to be weaker for more constrained dealers.

To test this effect, we relate our monetary policy pass-through measure to each dealer’s leverage ratio reporting during a given quarter. The results are reported in Table 7. Across all specifications, the coefficient on *LeverageRatio* is positive and significant, implying that if a dealer is more constrained (i.e., has a lower leverage ratio), its passthrough of DFR changes to the client sector is slower. Based on the most conservative specification in column (4) which also includes fixed effects for each government council, a one-standard deviation increase in the leverage ratio is associated with a 0.9% improvement in passthrough for when dealers trade repos with the same customer. We also confirm the findings of Ballensiefen et al. (2023) and Nguyen et al. (2023) as we see that repos backed by country collateral that is, on average less special (e.g., Spanish or Italian collateral), exhibit a higher passthrough.

6 Conclusion

We show that three key money market frictions — limited balance sheet space, collateral heterogeneity, and market power — affect the setting of haircuts. Our results have important implications for monetary policy, reducing its efficiency and raising concerns about market stability. While further research is needed to fully understand the implications for market stability, what is already evident

is that repo contracts with suboptimally chosen haircuts tend to exhibit greater volatility in both rates and volumes. At a broader level, this paper highlights another channel through which the supervisory function of a central bank influences its monetary policy role as post-crisis regulations, such as the leverage ratio introduced after 2008, may have unintended adverse consequences for both systemic risk and the effectiveness of monetary policy.

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A Robustness tests

The following table mirrors the results from Table 3 on the basis of all repo trades with a *non-zero* haircut. This alleviates the concern that pre-set haircuts fixed at zero, as is customary in the US, are driving the results. The table confirms that after excluding zero haircuts, the results become even stronger.

Table A.8: Do balance sheet constraints matter for haircuts?

The dependent variable is the volume-weighted average haircut. Banks are assigned to leverage terciles each quarter based on their reported leverage ratios, with tercile 1 indicating the most constrained institutions (i.e., lowest leverage ratios). The baseline is the third tercile, i.e., the group of banks with relatively high leverage ratios (and low balance sheet constraints). The variable *Post* equals one for transactions occurring after June 2021, and zero otherwise. The interaction terms capture the change in haircut behaviour for each leverage-constrained group following this date. Columns (1)–(3) report results for collateral-driven markets; columns (4)–(6) repeat the specifications using funding-driven markets. The sample includes repo transactions between January 2019 and June 2024, restricted to short maturities (O/N, T/N, S/N, 1W), where either the borrower (columns 1–3) or the lender (columns 4–6) is an ECB-identified dealer, and the counterparty is not. Haircuts are limited to transactions with valid collateral characteristics and where the collateral is issued by general government institutions (S13 sector). Standard errors clustered at the dealer-level are reported in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

	Haircut (%)					
	Collateral driven			Funding driven		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Tercile1</i> × <i>Post</i>	2.680*** (0.568)	2.010*** (0.280)	1.311*** (0.186)	−0.097 (0.657)	0.261 (0.510)	−0.056 (0.251)
<i>Tercile2</i> × <i>Post</i>	1.581*** (0.382)	1.586*** (0.366)	1.161*** (0.172)	0.016 (0.487)	0.164 (0.430)	0.082 (0.281)
<i>N</i>	19 690	19 690	19 690	39 940	39 940	39 940
<i>R</i> ²	0.42	0.68	0.77	0.18	0.43	0.61
FE: Coll. country	X	X	X	X	X	X
FE: Coll. Sector	X	X	X	X	X	X
FE: Client		X			X	
FE: Dealer × Client			X			X
FE: Month	X	X	X	X	X	X
FE: Leverage Tercile	X	X	X	X	X	X

The next chart is an alteration of Figure 3, where we do not estimate the distribution on the transaction level, but rather on the dealer-client-day aggregation level. The shape of the distribution look very similar, suggesting that there is low within-day heterogeneity and that haircuts are set on a relationship-day level. Haircut-setting is likely done via a standing agreement and less often repriced for each individual transaction. This is consistent with standing collateral agreements with periodic resets and batch processing of multiple transactions at the same terms.

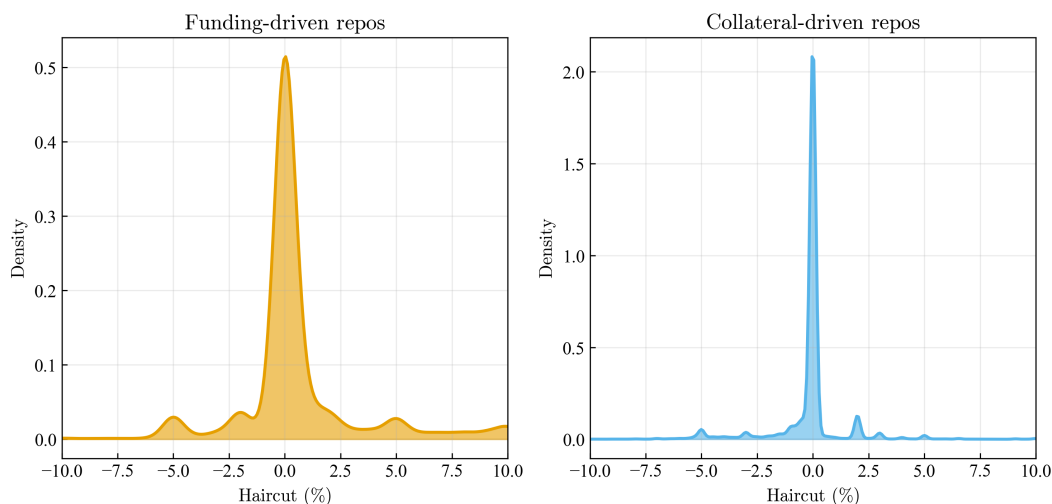


Figure A.8: What is the distribution of haircuts?

This histogram displays the estimated distribution of haircuts in funding-driven and collateral-driven repo transactions between dealer banks and non-dealer clients from January 2019 to June 2024 on a *dealer-client-day* level. The estimation used a Gaussian kernel.